

EduGenie – System Architecture

Project: EduGenie – Google Gemini Powered Learning Assistant

1. Introduction

The EduGenie system follows a simple web-based architecture. It consists of a frontend, a FastAPI backend, learning modules, and the Google Gemini AI service.

2. Architecture Components

2.1 User

The student interacts with EduGenie through a web browser and provides a question or learning request.

2.2 Frontend

The frontend is developed using HTML and CSS. It provides the user interface for selecting a learning task, entering information, and viewing the generated result.

2.3 FastAPI Backend

The FastAPI backend receives requests from the frontend, identifies the required learning operation, and sends the request to the appropriate module.

2.4 Learning Modules

EduGenie contains separate modules for different learning functions:

- **Q&A Module** – answers educational questions.
- **Explanation Module** – provides simplified explanations.
- **Quiz Module** – generates quiz questions.
- **Summary Module** – summarizes educational text.
- **Learning Path Module** – provides learning recommendations.

2.5 Google Gemini API

The Google Gemini API provides AI-powered processing for the required learning tasks and generates responses based on the user's input.

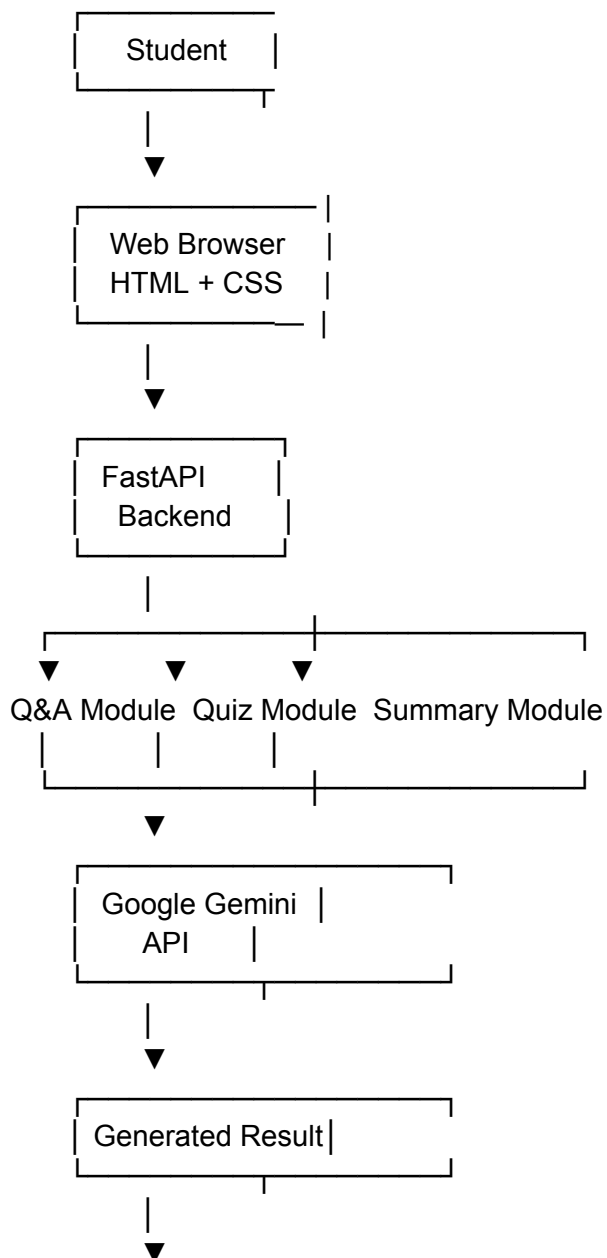
3. System Flow

The basic flow of the system is:

**Student → Web Interface → FastAPI Backend → Learning Module → Google Gemini API
→ Backend → Web Interface → Student**

4. Architecture Diagram

The architecture can be represented as:



5. Conclusion

The EduGenie architecture separates the user interface, backend processing, learning modules, and AI service. This structure makes the application easier to understand, maintain, and extend with additional learning features.